Azim Premji University

# MTH-113-2.00 - Introduction to Quantitative Thinking

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| **Course Name** | MTH-113-2.00 | **Course Title** | Introduction to Quantitative Thinking |
| **PS Course ID** | 669447 | **Base Program** | FXCRD-1.00 |
| **Course Version** | 2.00 | **Credits** | 5 |
| **Academic Year Version First Offered** | 2025 | **School** | School of Arts and Sciences |
| **Academic Year Version Last Offered** | | **Primary Course Author** | Achyut Kumar Banerjee |
| **Course Status** | Requested | **Course Review Status** | Under Creation |
| **Availability Outside Base Program** | Yes | **Base Program for Requested Course** | Flexible credit |
| **Requested Course Name** | Quantitative Reasoning: Introduction | **Total Assessment Percentage** | 100 |
| **Review Team** | | | |
| **CRC Team** | CRC-SAS-2025-2026 | **CRC Coordinator** | M S R Kumar |
| **CPC Team** | CPC-2025-2026 | **CPC Coordinator** | Ananya Banerjee |
| **AC Team** | AC-2025-2026 | **AC Coordinator** | Anand Shrivastava |
| **Review Timelines** | | | |
| **CRC Review Deadline** | 13/02/2026 | **CPC Review Deadline** | 13/03/2026 |
| **AC Review Deadline** | 20/03/2026 | | |
| **Curricular Group Information** | | | |
| **Curricular Group 1** | Flexible Credits | **Curricular Group 2** | |
| **Curricular Group 3** | | **Curricular Group 4** | |
| **Curricular Group 5** | | | |
| **Course Type Information** | | | |
| **Course Type 1** | In person on campus | **Course Type 2** | |
| **Course Type 3** | | **Course Type 4** | Classroom instruction |
| **Course Type 5** | | | |
| **Pre-requisite and Eligibility** | | | |
| **Prerequisites Course 1** | | **Prerequisites Course 2** | |
| **Prerequisites Course 3** | | **Prerequisites Course 4** | |
| **Prerequisites Course 5** | | **Eligibility Criteria** | |
| **Any Other Exclusion** | | | |
| **Other Details** | | | |
| **Course delivery mode** | Offline | **Level** | 1 |
| **Number Of Teaching Hours** | 60 | | |

| **UDL Details** | |
|---|---|
| **Alignment to UDL** | No |
| **Description of UDL Alignment** | |
| **UDL Alignment** | Click here to view the uploaded document !! |

**Development Team**

**Author**

| **Author's Name** | **Role** |
|---|---|
| Achyut Kumar Banerjee | Author |

**Co-Author**

| **Co-Author's Name** | **Role** | **Co-Author Email** | **Organization** |
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**Abstract Description**

This foundational course aims to introduce students to the core components of quantitative reasoning and mathematical methods, equip them with practical data skills, including the ability to interpret graphs, calculate statistical measures, and use spreadsheet software effectively to solve problems. The course does not have any prerequisites; rather, it intends to impart foundational knowledge of mathematical thinking that is expected to help students in other courses across disciplines.

**Description**

Introduction to Quantitative Thinking is a six-week summer bridge course designed exclusively for first-year undergraduate students. This support course meets the critical need for mathematical preparation, enabling all students, regardless of their prior mathematical background, to engage successfully with the core and the common curriculum courses and choose any academic pathway they wish to pursue.
The course will be delivered over six weeks during the summer program, immediately before the first semester begins. Total instruction consists of 60 hours of intensive classroom teaching conducted over four days each week, allowing for personalized support and peer collaboration. This concentrated, immersive format ensures that students build competency rapidly while remaining engaged and supported throughout their learning journey.

**Rationale and Introduction**

Diversity and inclusion are central to our university. For many of our students, a lack of preparation (especially in Mathematics and English) has been an obstacle to success. This course, along with the language course, is a formalization of the support system aimed at equipping our students with the necessary quantitative skills to navigate the core and common curriculum courses.
As part of the common curriculum in the undergraduate program, every student has to take the courses Public Reasoning (PRE) and The Measures of India (Understanding India-III). The former course requires some experience in logical reasoning, and the latter requires some statistical skills. In addition, it would be ideal to empower our students with the capability to choose any occupational track, including ‘Data, Development, and Democracy’, which requires solid quantitative understanding.
This is, therefore, a support course designed for students who may benefit from additional assistance in developing their foundational knowledge in mathematics and statistics, particularly those who may not have yet fully demonstrated proficiency during the admission process, to successfully navigate the core and common curriculum courses with mathematics and statistics content.

| **Intended Learning Outcomes** | |
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| **Intended Learning Outcome** | **Description** |
| Demonstrate | Demonstrate foundational knowledge of arithmetic operations, exponents, algebraic expressions, and coordinate geometry |
| Engage | Engage in mathematical thinking and collaborative problem-solving to build confidence and quantitative literacy. |
| Execute | Execute statistical and probabilistic reasoning, estimation methods, and data visualization to solve real world problems. |
| Perform | Carry out mathematical operations and problem-solving techniques in numerical, spatial, and data analysis contexts. |

| **Pedagogy** | |
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| **Pedagogical Practice** | **Pedagogy Description** |
| Lecture | Concise presentations followed by guided practice and problem-solving to reinforce learning. |
| Problem based learning | Students solve carefully designed problems individually and in groups with instructor support. |
| Project-based learning | Students apply multi-unit concepts through real-world collaborative projects involving data management, visualization, and interpretation. |

| **Assessment** | | | |
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| **Total Assessment Percentage** | | 100 | |
| **Assessment Type** | **Number of ILO Linkage** | **Assessment Percentage** | **Description** |
| In-class exam | 2 | 10 | Exam 4 (60 minutes duration, to be submitted individually), to be conducted in 6th week, will be a pen and paper-based assessment on the syllabus of unit 4. The assessment will include a mix of question types (multiple-choice and short problem-solving questions) and will evaluate students' ability to apply quantitative knowledge in everyday life. |
| In-class exam | 2 | 10 | Exam 2 (60 minutes duration, to be submitted individually), to be conducted in 3rd week, will be a pen and paper-based assessment on the syllabus of unit 2. The assessment will include a mix of question types (multiple-choice and short problem-solving questions) and will evaluate students' ability to understand algebra and coordinate geometry. |
| In-class exam | 1 | 10 | Exam 3 (60 minutes duration, to be submitted individually), to be conducted in 4th week, will be a computer-based assessment on the syllabus of Unit 3. The assessment will evaluate students' ability to estimate statistical measures in a spreadsheet program. |

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| In-class exam | 2 | 10 | Exam 1 (60 minutes duration, to be submitted individually), to be conducted in 2nd week, will be a pen and paper-based assessment on the syllabus of unit 1. The assessment will include a mix of question types (multiple-choice and short problem-solving questions) and will evaluate students' foundational understanding of the quantitative methods. |
| Take home exam | 2 | 20 | Two take home assignments (to be submitted individually) – one on the syllabi of units 1 and 2 (3rd week), and another on the syllabus of unit 4 (6th week). The assessment will include short problem-solving questions and will evaluate students' foundational understanding of the quantitative methods. |
| Project | 2 | 30 | The students will be divided into groups. Each group will work on a project related to an everyday life issue ( for example, breakfast habit, reading time). This will involve data acquisition (from small surveys), processing, and visualization. The project will start in the 3rd week of the course. Each group will present their project findings as a digital/poster presentation. The presentation will carry 10% weightage. It will be evaluated based on topic selection and presentation skill (including visual appeal and interaction). Individual members must indicate their roles in the project. Individual student will submit a brief report, which will contribute 10% to the overall grade. The report will be evaluated based on its organization and information of the workflow (data collection, analysis, and interpretation of the findings). The viva-voce will evaluate contribution and depth of knowledge of individual group members and will contribute 10% to the total grade. In the viva-voce, the instructor may ask the students to demonstrate use of a specific tool or function of the software related to their projects. The instructor should ensure that the groups are heterogeneous with students across all disciplines. The teams should not be larger than 5 and not smaller than 2. All projects will be discussed a priori. The students should start working on the projects from week 3. The presentation, report submission, and viva-voce will be conducted in week 5. |
| Class participation | 4 | 10 | This will be based on the students' interactions with peers and their contribution to discussions in the class. The students will be divided into groups and engaged in different types of in-class activities. The instructor may use this assessment to gather continuous feedback from the students, including their challenges in navigating the course, and take the necessary actions. |

**Syllabus Summary**

The course is structured into four progressive units that build quantitative reasoning from fundamentals to application.

- Unit 1 establishes essential building blocks through arithmetic operations, factorization, exponents and powers, and the unitary method for measurement. (Week 1)
- Unit 2 develops algebraic thinking and spatial representation through linear, polynomial, and quadratic relationships, and their graphical representations using coordinate geometry. (Weeks 2-3)
- Unit 3 builds data competency through an introduction to statistics, data visualization, interpretation, and understanding data distributions. (Weeks 3-4)
- Unit 4 extends quantitative reasoning to numerical applications, including percentage and compound interest, ratio and proportion, permutations and combinations, sequences and series, and probability. (Weeks 5-6)

The key resources for this course are listed below:
Textbooks:
1. Textbooks (VI – XI) by *NCERT*
2. NCERT Exemplar Problems

## Syllabus

**Unit Name**
Unit 1: Foundation for Quantitative Methods

**Description**
Position the unit to build essential building blocks for more advanced topics.

Week 1:
1.1 Number System
1.2 Arithmetic Operations
1.3 Prime Factorization (LCM and HCF)
1.4 Exponents and Power
1.5 Unitary Method

**Readings**
**Core Readings:**
NCERT Class VI, Chapters:

- Playing with Numbers

- Integers
- Fractions
- Decimals

NCERT Class VII, Chapters:

- Integers
- Fractions and Decimals
- Exponents and Powers

NCERT Class VIII, Chapter:

- Exponents and Powers

All core readings for this chapter are uploaded **here.**

**Supplementary Readings:**
NCERT Class VII, Chapter: Rational Numbers
Specifics Units of:
NCERT Exemplar Class VI
NCERT Exemplar Class VII
NCERT Exemplar Class VIII

**Unit Name**
Unit 2: Algebra and Coordinate Geometry

**Description**
Algebra is a useful tool to convert the problem into solving a system of linear equations. Solutions to linear equations also arise in several real-life applications. Graphs of linear functions will also be discussed.

Week 2:
2.1 Introduction to Algebra
2.2 Algebraic Expression
2.3 Factorization
2.4 Algebraic Identities
2.5 Polynomials

Week 3:
2.6 Linear Equations
2.7 Quadratic Equation
2.8 Coordinate Geometry
2.9 Graphs of Linear Equations

**Readings**
**Core Readings:**
NCERT Class VI, Chapter:

- Algebra

NCERT Class VII, Chapters:

- Algebraic Expression
- Simple Equation

NCERT Class VIII, Chapters:

- Algebraic Expression and Identities
- Factorization
- Linear Equation in One Variable

NCERT Class IX, Chapters:

- Linear Equation in Two Variables
- Polynomials
- Coordinate Geometry

NCERT Class X, Chapters:

- Linear Equation in Two Variables
- Polynomials
- Quadratic Equations

All core readings for this chapter are uploaded **here.**

**Supplementary Readings:**
Specific Units of:
NCERT Exemplar Class VI
NCERT Exemplar Class VII
NCERT Exemplar Class VIII
NCERT Exemplar Class IX
NCERT Exemplar Class X

**Unit Name**
Unit 3: Statistics, Data Management and Visualization

**Description**
Plotting and interpreting graphs is of considerable importance as a life skill. Line graphs, pie and bar charts, organization of quantitative data, measures of central tendency, and measures of variance will be covered in this unit. Students will also be introduced to the use of software (MS Excel or Google Sheets). They will be required to use the software to enter simple formulas and use it to calculate statistical measures.

Week 3:
3.1 Introduction to Statistics
i) Measures of Central Tendency
ii) Measures of Variance

Week 4:
3.2 Normal Distribution
3.3 Data Handling and Plotting Graphs
3.4 Data Visualization and Interpretation using Google Spreadsheet or MS Excel

**Readings**
**Core Readings:**
NCERT Class VII, Chapter:

- Data Handling

NCERT Class VIII, Chapter:

- Introduction to Graphs

NCERT Class IX, Chapter:

- Statistics

NCERT Class X, Chapter:

- Statistics

NCERT Class XI, Chapter:

- Statistics

All core readings for this chapter are uploaded **here.**

**Supplementary Readings:**
NCERT Exemplar Class IX
NCERT Exemplar Class X
NCERT Exemplar Class XI

**Unit Name**
Unit 4: Numerical Reasoning and Probability

**Description**

This unit builds on the first unit by using arithmetic to understand crucial ideas applicable in everyday life. This includes working with ratios and proportions, percentages, simple and compound interest, etc. Focus will be on the simplest of relations when two quantities are related linearly.
The unit will also touch upon some important concepts like progression, permutation, combination, and probability.

Week 5:
4.1 Percentage
4.2 Ratio and Proportion
i) Direct Proportion
ii) Inverse Proportion
4.3 Arithmetic Progression

Week 6:
4.4 Permutation and Combination
4.5 Probability

**Readings**
**Core Readings:**
NCERT Class VI, Chapter:

- Ratio and Proportion

NCERT Class VII, Chapter:

- Comparing Quantities

NCERT Class VIII, Chapters:

- Comparing Quantities
- Direct and Inverse Proportions

NCERT Class IX, Chapter:

- Probability

NCERT Class X, Chapters:

- Arithmetic Progression
- Probability

NCERT Class XI, Chapters:

- Probability
- Permutation and Combination

All core readings for this chapter are uploaded **here.**

***Supplementary Readings:***
NCERT Exemplar Class IX
NCERT Exemplar Class X
NCERT Exemplar Class XI

## Annexure

**ILO Assessment Linkage**

| ILO | Assessment | Assessment Notes |
|---|---|---|
| | In-class exam<br>In-class exam | Exam 1 (60 minutes duration, to be submitted individually), to be conducted in 2nd week, will be a pen and paper-based assessment on the syllabus of unit 1. The assessment will include a mix of question types (multiple-choice and short problem-solving questions) and will evaluate students' foundational understanding of the quantitative methods.<br><br>Exam 2 (60 minutes duration, to be submitted individually), to be conducted in 3rd week, will be a pen and paper-based assessment on the syllabus of unit 2. The assessment will include a mix of question types (multiple-choice and short problem-solving questions) and will evaluate students' ability to understand algebra and coordinate geometry. |

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| Demonstrate | Take home exam<br><br>Class participation | Two take home assignments (to be submitted individually) – one on the syllabi of units 1 and 2 (3rd week), and another on the syllabus of unit 4 (6th week). The assessment will include short problem-solving questions and will evaluate students' foundational understanding of the quantitative methods.<br><br>This will be based on the students' interactions with peers and their contribution to discussions in the class. The students will be divided into groups and engaged in different types of in-class activities. The instructor may use this assessment to gather continuous feedback from the students, including their challenges in navigating the course, and take the necessary actions. |
| Engage | In-class exam<br><br>Project<br><br>Class participation | Exam 4 (60 minutes duration, to be submitted individually), to be conducted in 6th week, will be a pen and paper-based assessment on the syllabus of unit 4. The assessment will include a mix of question types (multiple-choice and short problem-solving questions) and will evaluate students' ability to apply quantitative knowledge in everyday life.<br><br>The students will be divided into groups. Each group will work on a project related to an everyday life issue ( for example, breakfast habit, reading time). This will involve data acquisition (from small surveys), processing, and visualization. The project will start in the 3rd week of the course. Each group will present their project findings as a digital/poster presentation. The presentation will carry 10% weightage. It will be evaluated based on topic selection and presentation skill (including visual appeal and interaction). Individual members must indicate their roles in the project. Individual student will submit a brief report, which will contribute 10% to the overall grade. The report will be evaluated based on its organization and information of the workflow (data collection, analysis, and interpretation of the findings). The viva-voce will evaluate contribution and depth of knowledge of individual group members and will contribute 10% to the total grade. In the viva-voce, the instructor may ask the students to demonstrate use of a specific tool or function of the software related to their projects. The instructor should ensure that the groups are heterogeneous with students across all disciplines. The teams should not be larger than 5 and not smaller than 2. All projects will be discussed a priori. The students should start working on the projects from week 3. The presentation, report submission, and viva-voce will be conducted in week 5.<br><br>This will be based on the students' interactions with peers and their contribution to discussions in the class. The students will be divided into groups and engaged in different types of in-class activities. The instructor may use this assessment to gather continuous feedback from the students, including their challenges in navigating the course, and take the necessary actions. |
| Execute | In-class exam<br><br>Project<br><br>Class participation | Exam 3 (60 minutes duration, to be submitted individually), to be conducted in 4th week, will be a computer-based assessment on the syllabus of Unit 3. The assessment will evaluate students' ability to estimate statistical measures in a spreadsheet program.<br><br>The students will be divided into groups. Each group will work on a project related to an everyday life issue ( for example, breakfast habit, reading time). This will involve data acquisition (from small surveys), processing, and visualization. The project will start in the 3rd week of the course. Each group will present their project findings as a digital/poster presentation. The presentation will carry 10% weightage. It will be evaluated based on topic selection and presentation skill (including visual appeal and interaction). Individual members must indicate their roles in the project. Individual student will submit a brief report, which will contribute 10% to the overall grade. The report will be evaluated based on its organization and information of the workflow (data collection, analysis, and interpretation of the findings). The viva-voce will evaluate contribution and depth of knowledge of individual group members and will contribute 10% to the total grade. In the viva-voce, the instructor may ask the students to demonstrate use of a specific tool or function of the software related to their projects. The instructor should ensure that the groups are heterogeneous with students across all disciplines. The teams should not be larger than 5 and not smaller than 2. All projects will be discussed a priori. The students should start working on the projects from week 3. The presentation, report submission, and viva-voce will be conducted in week 5.<br><br>This will be based on the students' interactions with peers and their contribution to discussions in the class. The students will be divided into groups and engaged in different types of in-class activities. The instructor may use this assessment to gather continuous feedback from the students, including their challenges in navigating the course, and take the necessary actions. |
| | | Exam 1 (60 minutes duration, to be submitted individually), to be conducted in 2nd week, will be a pen and paper-based assessment on the syllabus of unit 1. The assessment will include a mix of question types (multiple-choice and short problem-solving questions) and will evaluate students' foundational understanding |

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| Perform | In-class exam<br><br>In-class exam<br><br>In-class exam<br><br>Take home exam<br><br>Class participation | of the quantitative methods.<br><br>Exam 2 (60 minutes duration, to be submitted individually), to be conducted in 3rd week, will be a pen and paper-based assessment on the syllabus of unit 2. The assessment will include a mix of question types (multiple-choice and short problem-solving questions) and will evaluate students' ability to understand algebra and coordinate geometry.<br><br>Exam 4 (60 minutes duration, to be submitted individually), to be conducted in 6th week, will be a pen and paper-based assessment on the syllabus of unit 4. The assessment will include a mix of question types (multiple-choice and short problem-solving questions) and will evaluate students' ability to apply quantitative knowledge in everyday life.<br><br>Two take home assignments (to be submitted individually) – one on the syllabi of units 1 and 2 (3rd week), and another on the syllabus of unit 4 (6th week). The assessment will include short problem-solving questions and will evaluate students' foundational understanding of the quantitative methods.<br><br>This will be based on the students' interactions with peers and their contribution to discussions in the class. The students will be divided into groups and engaged in different types of in-class activities. The instructor may use this assessment to gather continuous feedback from the students, including their challenges in navigating the course, and take the necessary actions. |

**Assessment ILO Linkage**

| Assessment | Assessment Notes | ILO |
|---|---|---|
| Class participation | This will be based on the students' interactions with peers and their contribution to discussions in the class. The students will be divided into groups and engaged in different types of in-class activities. The instructor may use this assessment to gather continuous feedback from the students, including their challenges in navigating the course, and take the necessary actions. | Demonstrate<br><br>Perform<br><br>Execute<br><br>Engage |
| In-class exam | Exam 2 (60 minutes duration, to be submitted individually), to be conducted in 3rd week, will be a pen and paper-based assessment on the syllabus of unit 2. The assessment will include a mix of question types (multiple-choice and short problem-solving questions) and will evaluate students' ability to understand algebra and coordinate geometry. | Demonstrate<br><br>Perform |
| In-class exam | Exam 3 (60 minutes duration, to be submitted individually), to be conducted in 4th week, will be a computer-based assessment on the syllabus of Unit 3. The assessment will evaluate students' ability to estimate statistical measures in a spreadsheet program. | Execute |
| In-class exam | Exam 1 (60 minutes duration, to be submitted individually), to be conducted in 2nd week, will be a pen and paper-based assessment on the syllabus of unit 1. The assessment will include a mix of question types (multiple-choice and short problem-solving questions) and will evaluate students' foundational understanding of the quantitative methods. | Demonstrate<br><br>Perform |
| In-class exam | Exam 4 (60 minutes duration, to be submitted individually), to be conducted in 6th week, will be a pen and paper-based assessment on the syllabus of unit 4. The assessment will include a mix of question types (multiple-choice and short problem-solving questions) and will evaluate students' ability to apply quantitative knowledge in everyday life. | Perform<br><br>Engage |
| | The students will be divided into groups. Each group will work on a project related to an everyday life issue ( for example, breakfast habit, reading time). This will involve data acquisition (from small surveys), processing, and visualization. The project will start in the 3rd week of the course. Each group will | |

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| Project | present their project findings as a digital/poster presentation. The presentation will carry 10% weightage. It will be evaluated based on topic selection and presentation skill (including visual appeal and interaction). Individual members must indicate their roles in the project. Individual student will submit a brief report, which will contribute 10% to the overall grade. The report will be evaluated based on its organization and information of the workflow (data collection, analysis, and interpretation of the findings). The viva-voce will evaluate contribution and depth of knowledge of individual group members and will contribute 10% to the total grade. In the viva-voce, the instructor may ask the students to demonstrate use of a specific tool or function of the software related to their projects. The instructor should ensure that the groups are heterogeneous with students across all disciplines. The teams should not be larger than 5 and not smaller than 2. All projects will be discussed a priori. The students should start working on the projects from week 3. The presentation, report submission, and viva-voce will be conducted in week 5. | Execute<br><br>Engage |
| Take home exam | Two take home assignments (to be submitted individually) – one on the syllabi of units 1 and 2 (3rd week), and another on the syllabus of unit 4 (6th week). The assessment will include short problem-solving questions and will evaluate students' foundational understanding of the quantitative methods. | Demonstrate<br><br>Perform |

| Course Review Feedback | | | | |
|---|---|---|---|---|
| **Type** | **Section** | **Status** | **Review/Overall Comments** | **Authors Comments** |
| CRC | Assessment | | I was wondering about the use of AI in the two take-home assignments, which together amount to 20% of the assessment weightage. If the intention is for these to be strictly non-AI, would it be possible to convert them into an in-class open-book exams where students are allowed to refer to their notes? Alternatively, could a short post-submission oral viva be included as part of the assessment, which may also serve as a deterrent to AI usage? | |

| Reason for Revision | |
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| **Reason for Revision** | Instructor Feedback |
| **Reason (Descriptive)** | Assessment: Complete Change in Assessment Type, Syllabus Change |
| **Date of Revision** | 01/21/2026 |
| **Course Created from** | MTH-113-1.00 |

| Revision Summary | |
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| **Overall Revision Summary** | This is a previously approved course. But for consistency, the title has been changed to: *Quantitative Reasoning: Introduction* The following revisions are made. |
| **Summary of Revision in Syllabus** | The course is structured into four progressive units that build quantitative reasoning from fundamentals to application. Unit 1 establishes essential building blocks through arithmetic operations, factorization, exponents and powers, and the unitary method for measurement (Week 1). Unit 2 develops algebraic thinking and spatial representation through linear, polynomial, and quadratic relationships, and their graphical representations using coordinate geometry (Weeks 2-3). Unit 3 builds data competency through an introduction to statistics, data visualization, and interpretation, and understanding data distributions (Weeks 3-4). Unit 4 extends quantitative reasoning to numerical applications, including percentage and compound interest, ratio and proportion, permutations and combinations, sequences and series, and probability (Weeks 5-6). |
| **Summary of Revision in ILO** | ILO 1: Demonstrate (Demonstrate foundational knowledge of arithmetic operations, exponents, algebraic expressions, and coordinate geometry)<br>ILO 2: Perform (Carry out mathematical operations and problem-solving techniques in numerical, spatial, and data analysis contexts) |

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| | ILO 3: Execute (Execute statistical and probabilistic reasoning, estimation methods, and data visualization to solve real world problems)<br>ILO 4: Engage (Engage in mathematical thinking and collaborative problem-solving to build confidence and quantitative literacy) |
| **Summary of Revision in Pedagogy** | Lecture (Concise presentations followed by guided practice and problem-solving to reinforce learning)<br>Problem-based learning (Students solve carefully designed problems individually and in groups with instructor support)<br>Project-based learning (Students apply multi-unit concepts through real-world collaborative projects involving data management, visualization a,nd interpretation ) |
| **Summary of Revision in Assessment** | 4 in-class exams, 1 project, 2 take-home assignments, and class participation |